Samrah Sayyed

Business Analytics • AI Applications • Embedded Systems Research

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PROFESSIONAL SUMMARY

Electrical and Computer Engineering undergraduate with hands-on experience in Business Analytics, Full-Stack Development, AI-powered applications, and Embedded Systems research. Skilled in SQL, Python, Power BI, React.js, and STM32, with experience building data-driven solutions, scalable software systems, and real-time automotive communication platforms. Published researcher with strong analytical and problem-solving abilities and a passion for leveraging data and technology to develop intelligent, impactful solutions.

EDUCATIONS

MIT WORLD PEACE UNIVERSITY | PUNE — Aug 2023 - May 2027

Bachelor's in Technology in Electrical and Computer Engineering — *CGPA : 8.10*

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Computer Organization and Operating Systems, Artificial Intelligence & Machine Learning, Data Science, Microcontroller and Applications, Embedded Systems, Communication Networks.

10th - CBSE Percentage: 89% 12th - HSC Percentage: 76%

PROJECTS

Pharmaceutical Commercial Analytics & Sales Intelligence Platform

Analyst | MySQL, Power BI, Excel, Python (Pandas), GitHub

• Designed and developed an end-to-end commercial analytics solution for a pharmaceutical company using a dataset of 10,000 sales records, 100 products, 50 sales representatives, and 200 physicians to evaluate sales performance and commercial effectiveness.
• Performed SQL-based data extraction, cleaning, aggregation, and analysis to identify product performance, regional sales trends, physician segmentation, campaign ROI, and sales representative effectiveness.
• Built four interactive Power BI dashboards - Executive Overview, Product & Sales Analytics, Doctor & Representative Performance, and Marketing Campaign Analytics, to monitor 10+ KPIs, including revenue growth, prescription trends, conversion rates, market share, and representative productivity.
• Generated data-driven recommendations to optimize marketing investments, improve territory planning and physician targeting, and enhance overall commercial performance and decision-making.
• Gathered business requirements, defined analytical KPIs, designed a relational data model, and documented stakeholder objectives and executive recommendations to simulate a real-world pharmaceutical consulting engagement.

AI-Powered Study Analytics & Productivity Platform | Github Website

Developer | React.js, Node.js, Express.js, MongoDB, AI APIs, Postman

• Developed an AI-powered productivity platform supporting study planning, task management, focus tracking, analytics dashboards, and personalized learning workflows using React.js, Node.js, Express.js, and MongoDB.
• Engineered scalable backend services, REST APIs, and secure authentication mechanisms to support user management, activity tracking, and seamless application performance.
• Integrated Generative AI and Retrieval-Augmented Generation (RAG) pipelines to provide personalized productivity insights, study recommendations, and academic assistance.
• Designed an analytics engine to capture and visualize study-session data, enabling users to identify productivity trends, subject-wise effort distribution, and learning consistency.
• Applied modular software engineering practices, API integrations, and Git-based version control to improve maintainability and support future scalability.

LiDAR and RADAR Sensor Fusion with STM32

Researcher | STM32, Embedded C, Unity, RPLiDAR, Radar Sensor, RealTerm

• Designed and implemented a real-time sensor fusion system integrating LiDAR and radar data for intelligent vehicle perception and Advanced Driver Assistance System (ADAS) applications.
• Developed obstacle detection and sensor-processing algorithms using STM32-based embedded systems and validated outputs through simulation and hardware testing.
• Extended CAN-based communication infrastructure to support reliable transmission of high-volume sensor data across distributed vehicle subsystems.
• Built a Unity-based 3D simulation environment for algorithm validation, troubleshooting, and performance optimization before hardware deployment.
• Established the foundation for future sensor fusion enhancements, including Time-to-Collision (TTC) estimation and brake-by-wire integration.

Integration of Sensor with ECU over CAN Protocol

Researcher | STM32, Embedded C, CAN Protocol

• Developed a distributed embedded communication framework using STM32 microcontrollers and CAN protocol for real-time sensor-to-ECU communication.
• Implemented fault-tolerant data transmission mechanisms to support scalable automotive system architectures and improve communication reliability.
• Performed testing, debugging, and validation of communication workflows to ensure robust real-time performance.
• Contributed to technical documentation and a conference publication on CAN-based multi-sensor communication systems and supported an invention disclosure currently under patent evaluation.

PROJECTS

SkillLink- Social Media Web Application | Github

Developer | React.js, Node.js, MySQL, Tailwind CSS

- Built a full-stack social networking application featuring authentication, user profiles, posts, comments, likes, and messaging functionality.
- Developed scalable RESTful APIs using Node.js and Express.js and integrated them with a React.js frontend.
- Designed and optimized MySQL database schemas and CRUD operations to improve application performance and data management.
- Implemented secure authentication, session management, and responsive UI design following software engineering best practices.

Home Automation System

Developer | Arduino IDE, Blynk IoT, Embedded Systems

- Developed an IoT-based smart home automation system for remote monitoring and control of household devices and environmental sensors.
- Integrated temperature, humidity, gas leakage, motion detection, and ambient light sensing with real-time monitoring capabilities.
- Implemented automation workflows, embedded control logic, and cloud-connected monitoring through Blynk IoT and an OLED feedback system.

TECHNICAL SKILLS

Programming Languages :	Python, C, C++, JavaScript, SQL, MATLAB
Data & Business Analytics :	SQL, Microsoft Excel, Power BI, Data Visualization, KPI Development, Dashboard Development, Business Intelligence, Requirement Analysis, Exploratory Data Analysis (EDA)
Software Development :	Data Structures & Algorithms, Object-Oriented Programming, Backend Development, Database Design, REST APIs, Authentication, Software Engineering
Frameworks & Technologies :	React.js, Node.js, Express.js, Tailwind CSS, HTML, CSS
Databases :	MySQL, MongoDB
AI & Cloud Technologies :	Generative AI, Retrieval-Augmented Generation (RAG), AI APIs, Prompt Engineering, Machine Learning Fundamentals, Cloud Computing Fundamentals, Docker (Learning)
Embedded Systems & Automotive :	STM32, Embedded C, CAN Protocol, IoT Systems, CANoe, CANalyzer, Vehicle Diagnostics, AUTOSAR (Fundamentals)
Tools & Platforms :	Git, GitHub, Postman, Google Colab, Microsoft PowerPoint, MATLAB Simulink, Stateflow
Languages :	English, Hindi, Marathi, French, Japanese

PUBLICATION, LEADERSHIP & ACHIEVEMENTS

- Co-author & POC, "Design and Implementation of Multi-Sensor Integration using STM32 over CAN-Based Communication Framework," published in IEEE Xplore, Jan 2026.
- Secured 2nd Place in the Research Poster Competition for "Digi Agri Cloud," presenting an innovative digital agriculture solution.
- Founding Member, The Robotics Society (TRS), MIT World Peace University, contributing to the establishment of the university's robotics community and supporting technical initiatives, workshops, and student engagement activities.
- Elected Student Representative for the Department of Electrical and Electronics Engineering, facilitating communication between students and faculty and supporting academic coordination initiatives.
- Served as Third-Year (TY) Coordinator and coordinated inter-departmental academic, sports, and cultural events under the Triangle Trophy program, managing logistics, scheduling, and team execution for 300+ students.
- Led the planning and execution of 5+ technical workshops, academic events, and student engagement initiatives through the Electrical Engineering Students' Association (EESA), driving participation across multiple student cohorts.
- Solved 100+ Data Structures and Algorithms problems across arrays, trees, graphs, dynamic programming, and greedy algorithms using C++ and Python.